PRODUCT USER MANUAL

For the Global Ocean Low and Mid Trophic Levels Biomass Content Hindcast Product

GLOBAL_MULTIYEAR_BGC_001_033
Issue: 4.4

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Approval Date : November 2025

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	**06/02/2019**	All	Initial version	Olivier Titaud	
1.1	**01/10/2019**	All	Extension with the year 2018	Olivier Titaud	
2.0	**04/05/2021**	All	New 1/12° daily product	Olivier Titaud	C. Derval
3.0	**13/09/2021**	All	Time extension	Olivier Titaud	C. Derval
4.0	**16/06/2022**	All	Quality Improvement, Updated file format of dataset	Olivier Titaud	Copernicus Marine Product Management
4.1	**31/03/2023**	All	Small corrections	Olivier Titaud	Copernicus Marine Product Management
4.2	**01/06/2023**	All	Time extension	Olivier Titaud	Copernicus Marine Product Management
4.3	**31/05/2024**	All	New computation grid and Time extension	Olivier Titaud	O. Le Galloudec
4.4	**30/05/2025**	All	Correction CF standard name variable	Olivier Titaud	O. Le Galloudec

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
SEAPODYM	Spatial Ecosystem And POpulation DYnamic Model
LMTL	Low and Mid-Trophic Level
GLO	Global
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
MDS	Marine Data Store

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service global ocean Low and Mid Trophic Levels (LMTL) biomass content hindcast (no data assimilation) **GLOBAL_MULTIYEAR_BGC_001_033**. It provides daily instantaneous fields from January 1998.

This product is defined on a standard regular 1/12-degree grid. The variables are located on the same grid points.

The product **GLOBAL_MULTIYEAR_BGC_001_033** is organized in two datasets:

- **cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-i:** primary dataset containing 2D daily instantaneous values of mass content of zooplankton (1 variable), mass content of micronekton (6 variables), net primary productivity (1 variable) and euphotic zone depth (1 variable)

- **cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-i:** secondary dataset containing 3D daily instantaneous values of physical forcing fields that have been used to produce the primary dataset: bottom depth of pelagic layers, vertical mean over pelagic layers of currents and temperature.

The product is published on the Copernicus Marine Service dissemination server after automatic and human quality controls. Product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF4 format and follow CF-7 convention. However, standard name for zooplankton and micronekton variables may be still under validation. As variables of the second dataset are very specific, they did not have been proposed to the CF convention.

The analysis and forecasting system is described in the Quality Information Document (QUID – see link in REFERENCES).

b) History of changes

Date	Description of changes and impacted product
2019/07/19	Initial ¼ degree weekly product covering latitudes from 80S to 65°N.and period from 1998-01-07 to 2016-12-21. A single dataset containing biomass content of meso zooplankton and 6 functional groups of micronekton
2019/12/03	Quality improvement (better parameter calibration) Extension of product up to 2018-12-19
2021/05/21	New daily 1/12 degree product covering latitudes from 80S up to 90N and period from 1998-01-01 to 2019-12-31 New secondary dataset containing forcing fields: (3D daily instantaneous values of physical forcing fields that have been used to produce the primary dataset: bottom depth of pelagic layers, vertical mean over pelagic layers of currents and temperature). Change of variable names Change of standard names Updated documentation (QUID and PUM)
2023/03/31	Correction of numerical transport scheme and update of diffusion coefficient
2023/12/31	Time extension
2024/12/31	New computation grid and time extension
2025/xxxx	Time Extension Correction dataset name Correction standard names of micronekton related variables for CF compliant

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	GLOBAL_MULTIYEAR_BGC_001_033
Geographical coverage	-80°N – 90°N
Variables	See full list of variables in table below
Product Type	Multi Year
Available time series	From 1998/01/01
Temporal resolution	Daily instantaneous
Horizontal resolution	1/12 degree
Number of vertical levels	3 pelagic layers with dynamical space dependent depths.
Format	NetCDF4

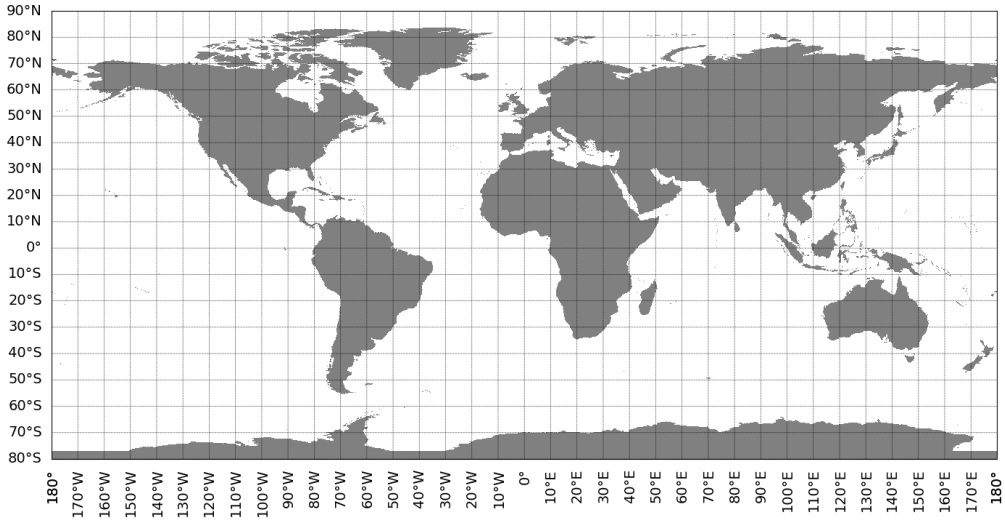
The production of GLOBAL_MULTIYEAR_BGC_001_033 provides global daily instantaneous hindcast fields of biomass content of zooplankton (one group) and micronekton (6 groups) at 1/12-degree spatial resolution.

Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid, and gelatinous species. It can be only a temporary stage for species with large growth potential.

Secondary variables and preprocessed forcing data that have been used to produce these fields are also delivered.

b) Production System Description

Summary of the operational model setups: Domain
Resolution and grid
Geographic coverage: global: -80°N - 90°N, -180°E - 180°E



The horizontal grid step is regular in latitude and longitude and is of 1/12° degree. All variables are collocated at the same point of this grid. Land-sea mask is adapted from the temperature and current land-sea mask.

Vertical grid	Pelagic layers.
System name	MICRORYSv2
Model version	SEAPODYM-LMTL_3.0
Assimilation scheme	N/A
Assimilated observations	None
Initial conditions	Climatological spin-up.
Forcings	Temperature and currents come from the Copernicus Marine Service Global Ocean Physics Reanalysis Satellite-based Net Primary Productivity is computed from the Copernicus Marine Service chlorophyll-a product (Global Ocean Chlorophyll from Satellite Observations), the Sea Surface Temperature (NOAA NCEI AVHRR product) and Surface Solar Radiation Downwards (ERA-INTERIM reanalysis). Missing NPP observations are completed by modelled NPP from the Copernicus Marine Service Global ocean biogeochemistry hindcast. Euphotic depth (used to compute the pelagic layers depths) is computed from chlorophyll a using Morel's case I model.

1. Description

Production of GLOBAL_MULTIYEAR_BGC_001_033 uses the Low and Mid-Trophic module of the Spatial Ecosystem And POpulation DYnamic Model (SEAPODYM-LMTL http://www.seapodym.eu/) (Lehodey et. al. 2008, 2010, 2015; Conchon 2015). It simulates the spatio-temporal dynamics of zooplankton and micronekton biomass density. All taxa are grouped together in so called functional groups depending on their position in the water column during day and night. Organisms inhabit three pelagic layers (namely epipelagic, upper mesopelagic and lower mesopelagic layers) whose depths vary in time and space. The zooplankton is modelled as a single functional group in the epipelagic layer. Micronekton includes fish, squid, and crustacean. It is modelled as six independent functional groups: 3 groups that always stay at a single layer and 3 groups that perform diel vertical migrations (i.e., they migrate between these layers during day and night alternation).

2. Forcings preprocessing and initial conditions

The model is forced by ocean currents, temperature, and Net Primary Productivity (NPP). Currents control the transport of organisms; temperature controls the time of development and NPP acts as a biomass source. Preprocessing starts with the computation of the depth of the three pelagic layers from euphotic depth. Then temperature and currents are averaged over these 3 layers. Depending on the diel vertical migration behavior of the considered functional group, temperature and currents are also averaged along the model time step over the layers it inhabits during day and night (weighted average with weights equal to the day/night duration ratio along the time step). Net Primary Productivity is computed from satellite-derived chlorophyl a, sea surface temperature and surface solar radiation downwards using the Vertically Generalized Production Model (VGPM; Behrenfeld and Falkowski (1997a)). When these observations are not available (i.e., at latitudes during boreal and austral winters), NPP comes from the Copernicus Marine Service biogeochemical reanalysis (PISCES model). A relaxation technique is used to smoothly adjust the data at the frontier of missing data. Initial conditions are obtained by climatological spin-ups. Advection-diffusion-reaction numerical computation is done in a Yin-Yang overset grids framework (Kageyama *et. al.* 2004). The pre-processing of forcing data uses GNU parallel (Tange 2021).

3. Population dynamics

In the core engine of the SEAPODYM model, zooplankton and micronecton populations are structured in discrete cohorts (subsample of the population that share the same age range). The number of cohorts depends on the maximal time of development of the organisms. At each time step, the first cohort is populated by a fraction of Net Primary Productivity. Each cohort is then advected by the currents and a mortality coefficient is applied. The GLOBAL_MULTIYEAR_BGC_001_033 product only contains the fraction of the considered functional group (integrated over all cohorts) that reached their maturity. This time of development (also called recruitment time) depends on the temperature organism experienced.

4. Variables units

Zooplankton biomass is expressed in carbon (g/m2). Micronekton is expressed in wet weight (g/m2). The factor used to convert carbon content to wet weight is delivered in NetCDF file as the variable attribute carbon_to_wet_weight_conversion_factor. For zooplankton this factor is set to 1. To get micronekton in gC the fields should be divided by this factor.

5. Vertical levels

Zooplankton and micronekton biomass content (primary dataset) is delivered at the three pelagic levels (namely epipelagic, upper and lower mesopelagic levels) depending on the considered functional groups. For functional groups that contain migrant organisms, the product is delivered at the level they inhabit at day. This later information is available as the variable attribute's day_layer and night_layer in NetCDF files.
Forcings (mainly in secondary dataset) are delivered at the epipelagic level for surface forcings (i.e. Net Primary Productivity and euphotic depth) and at the three pelagic levels for averaged temperature and currents.

6. Parameter calibration

Population dynamic parameters of the Low and Mid-Trophic Level module of SEAPODYM are based on results published in scientific literature (Lehodey et.al. 2010; 2015). See also the corresponding product QUality Information Document (QUID).

c) Processing information

7. Update Time

This product is disseminated in one go. It is homogeneous throughout the whole period.

8. Time averaging

The output fields are 1-day instantaneous integrated variables.

d) Details of datasets

GLOBAL_MULTIYEAR_BGC_001_033
Main dataset: cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-I Variables name in the NetCDF file and Unit: Description & Standard_name
zooc [g m-2] Mass content of zooplankton expressed in carbon. mass_content_of_zooplankton_expressed_as_carbon_in_sea_water
mnkc_epi [g m-2] Mass content of epipelagic micronekton expressed as wet weight. Component of the micronekton that inhabits permanently the epipelagic layer. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_epipelagic_micronekton
mnkc_umeso [g m-2] Mass content of upper mesopelagic micronekton expressed as wet weight. Component of the micronekton that inhabits permanently the upper mesopelagic layer. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_upper_mesopelagic_micronekton
mnkc_mumeso [g m-2] Mass content of migrant upper mesopelagic micronekton expressed as wet weight. Component of the micronekton that inhabits the upper mesopelagic layer during daytime and the epipelagic layer during nighttime. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid, and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_migrant_upper_mesopelagic_micronekton
mnkc_lmeso [g m-2] Mass content of lower mesopelagic micronekton expressed as wet weight. Component of the micronekton that inhabits permanently the lower mesopelagic layer. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_lower_mesopelagic_micronekton
mnkc_mlmeso [g m-2] Mass content of migrant lower mesopelagic micronekton expressed as wet weight. Component of the micronekton that inhabits the lower mesopelagic layer during daytime and the upper mesopelagic layer during nighttime. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid, and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_migrant_lower_mesopelagic_micronekton
mnkc_hmlmeso [g m-2] Mass content of highly migrant lower mesopelagic micronekton expressed as wet weight. Component of the micronekton that inhabits the lower mesopelagic layer during daytime

and the epipelagic layer during nighttime. Micronekton is classically defined by a size range of 2cm-20 cm. It contains a large diversity of fish, crustacean, squid, and gelatinous species. It can be only a temporary stage for species with large growth potential. ocean_wet_mass_content_of_highly_migrant_lower_mesopelagic_micronekton
npp [mg m-2 day-1] Net primary productivity of biomass net_primary_productivity_of_biomass_expressed_as_carbon_in_sea_water
zeu [m] Euphotic zone depth euphotic_zone_depth
Secondary dataset (physical forcings): cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-I Variables name in the NetCDF file and Unit: Description & Standard_name
pelagic_layer_depth [m] Depth of the bottom of pelagic layers considered in this product (3 layers). Pelagic layers are vertical regions of the water column of the open ocean that are inhabited by organisms. sea_water_pelagic_layer_bottom_depth
U [m s-1] Average from the bottom to the top of the considered pelagic layer of eastward current velocity field eastward_sea_water_velocity_vertical_mean_over_pelagic_layer
V [m s-1] Average from the bottom to the top of the considered pelagic layer of northward current velocity field northward_sea_water_velocity_vertical_mean_over_pelagic_layer
T [Celsius degree] Average from the bottom to the top of the considered pelagic layer of temperature field sea_water_potential_temperature_vertical_mean_over_pelagic_layer

e) Additional Information on parameters

pelagic_layer_depth [m]	This variable is a space-dependent dynamical 3D variable (time, latitude, longitude, depth) containing the bottom depths of the 3 pelagic layers the zooplankton and the micronekton inhabit. Dimension depth represents the index of the layers. The way this variable is computed is specified in NetCDF file and detailed in the QUID. It can be computed from euphotic depth or other environmental data.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " Files How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

a) Nomenclature of files when downloaded through the Copernicus Marine Web Portal Service

Files nomenclature when downloaded through the CMEMS MDS Services is based on the following nomenclature:
The scheme is: datasetname_nnnnnn.nc where:

- **datasetname** is a character string: cmems_mod_glo_bgc_my_0.083deg-lmtl_PT1D-I or cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-I
- **nnnnnn**: integer corresponding to the unique request number in MOTU.
- **nc**: standard NetCDF filename extension.

b) Nomenclature of original files

The filename scheme is datasetname_yyyymmdd.nc where

- **datasetname** is a character string: cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-I or cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-I
- **yyyymmdd**: field date string in "year-month-day" format. Year is specified with 4 digits. Month and day are specified with 2 digits (that is with eventually leading zeros). Example: 20190818.
- **nc**: standard NetCDF filename extension.

c) Other information: land mask value, compression

The NetCDF4 format is used with short integer coding for best compression, using an offset and scale factor as shown below:

Real_Value = (Display_Value X scale_factor) + add_offset
Fill values (correspond to land values): -32767s

Remark that due to the distribution of the variables in GLOBAL_MULTIYEAR_BGC_001_033 product, and in order to minimize the loss of information during this compression, scale_factor and add_offset values are not constant along the time series. For more information, please follow this link : How to use add_offset and scale_factor values ?

d) File size

GLOBAL_MULTIYEAR_BGC_001_033 dataset	Approximate size of an individual file (Mo)
cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-I	40
cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-I	90

d) Structure of files

For GLOBAL_MULTIYEAR_BGC_001_033 product:
Examples of the header of output NetCDF files are inserted in ANNEX, for each dataset.

5) REFERENCES

Behrenfeld, M. J. and Falkowski, P. G. (1997). Photosynthetic rates derived from satellite-based chlorophyll concentration. *Limnology and oceanography,* 42(1):1–20.

Conchon A. (2015). Modélisation du zooplancton et du micronecton marins. Sciences agricoles. Université de La Rochelle, 2016. Français. ⟨NNT : 2016LAROS011⟩. ⟨tel-01661554⟩

Kageyama, A. and Sato T. (2004). ''Yin-Yang grid'': An overset grid in spherical geometry, Geochem. Geophys. Geosyst., 5, Q09005, https://doi.org/10.1029/2004GC000734

Lehodey P., Murtugudde R., Senina I. (2010). Bridging the gap from ocean models to population dynamics of large marine predators: a model of mid-trophic functional groups. *Progress in Oceanography*, 84: 69–84.

Lehodey P., Senina I., Murtugudde R. (2008). A Spatial Ecosystem And Populations Dynamics Model (SEAPODYM) - Modelling of tuna and tuna-like populations. *Progress in Oceanography*, 78: 304-318.

Lehodey, P., Conchon, A., Senina, I., Domokos, R., Calmettes, B., Jouanno, J., Hernandez, O., and Kloser, R. (2015) Optimization of a micronekton model with acoustic data. – *ICES Journal of Marine Science*, 72(5): 1399-1412.

Tange, O. (2021) GNU Parallel 20210822 ('Kabul'). Zenodo. https://zenodo.org/doi/10.5281/zenodo.5233952

Quality Information Document (QUID) : CMEMS-GLO-QUID-001-033.pdf (copernicus.eu)

ANNEX

For GLOBAL_MULTIYEAR_BGC_001_033 product:

Examples of the header of output NetCDF files are inserted below, for each dataset.

For the cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-i dataset:

```
netcdf cmems_mod_glo_bgc_my_0.083deg-lmtl_P1D-i_20191225 {
dimensions:
        latitude = 2040 ;
        longitude = 4320 ;
        time = UNLIMITED ; // (1 currently)
variables:
        float latitude(latitude) ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        int time(time) ;
                time:standard_name = "time" ;
                time:units = "seconds since 1970-01-01 00:00:00" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
        short zooc(time, latitude, longitude) ;
                zooc:_FillValue = -32767s ;
                zooc:standard_name =
"mass_content_of_zooplankton_expressed_as_carbon_in_sea_water" ;
                zooc:units = "g m-2" ;
                zooc:scale_factor = 0.00547476163643558 ;
                zooc:add_offset = 179.386042831962 ;
        short mnkc_epi(time, latitude, longitude) ;
                mnkc_epi:_FillValue = -32767s ;
                mnkc_epi:standard_name =
"ocean_wet_mass_content_of_epipelagic_micronekton " ;
                mnkc_epi:units = "g m-2" ;
                mnkc_epi:scale_factor = 0.0882488280388828 ;
                mnkc_epi:add_offset = 2891.56110086868 ;
                mnkc_epi:day_layer = "epipelagic (1)" ;
                mnkc_epi:night_layer = "epipelagic (1)" ;
                mnkc_epi:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short mnkc_umeso(time, latitude, longitude) ;
                mnkc_umeso:_FillValue = -32767s ;
                mnkc_umeso:standard_name =
"ocean_wet_mass_content_of_upper_mesopelagic_micronekton " ;
                mnkc_umeso:units = "g m-2" ;
                mnkc_umeso:scale_factor = 0.0728645300670663 ;
                mnkc_umeso:add_offset = 2387.47919329244 ;
                mnkc_umeso:day_layer = "upper mesopelagic (2)" ;
```

```
                mnkc_umeso:night_layer = "upper mesopelagic (2)" ;
                mnkc_umeso:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short mnkc_mumeso(time, latitude, longitude) ;
                mnkc_mumeso:_FillValue = -32767s ;
                mnkc_mumeso:standard_name =
"ocean_wet_mass_content_of_migrant_upper_mesopelagic_micronekton " ;
                mnkc_mumeso:units = "g m-2" ;
                mnkc_mumeso:scale_factor = 0.292964517808274 ;
                mnkc_mumeso:add_offset = 9599.27539497627 ;
                mnkc_mumeso:day_layer = "upper mesopelagic (2)" ;
                mnkc_mumeso:night_layer = "epipelagic (1)" ;
                mnkc_mumeso:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short mnkc_lmeso(time, latitude, longitude) ;
                mnkc_lmeso:_FillValue = -32767s ;
                mnkc_lmeso:standard_name =
"ocean_wet_mass_content_of_lower_mesopelagic_micronekton " ;
                mnkc_lmeso:units = "g m-2" ;
                mnkc_lmeso:scale_factor = 0.0158402198688341 ;
                mnkc_lmeso:add_offset = 519.020644464163 ;
                mnkc_lmeso:day_layer = "lower mesopelagic (3)" ;
                mnkc_lmeso:night_layer = "lower mesopelagic (3)" ;
                mnkc_lmeso:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short mnkc_mlmeso(time, latitude, longitude) ;
                mnkc_mlmeso:_FillValue = -32767s ;
                mnkc_mlmeso:standard_name =
"ocean_wet_mass_content_of_migrant_lower_mesopelagic_micronekton_expressed " ;
                mnkc_mlmeso:units = "g m-2" ;
                mnkc_mlmeso:scale_factor = 0.0421403229894233 ;
                mnkc_mlmeso:add_offset = 1380.76982371494 ;
                mnkc_mlmeso:day_layer = "lower mesopelagic (3)" ;
                mnkc_mlmeso:night_layer = "upper mesopelagic (2)" ;
                mnkc_mlmeso:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short mnkc_hmlmeso(time, latitude, longitude) ;
                mnkc_hmlmeso:_FillValue = -32767s ;
                mnkc_hmlmeso:standard_name =
"ocean_wet_mass_content_of_highly_migrant_lower_mesopelagic_micronekton " ;
                mnkc_hmlmeso:units = "g m-2" ;
                mnkc_hmlmeso:scale_factor = 0.0574202298231246 ;
                mnkc_hmlmeso:add_offset = 1881.43125126068 ;
                mnkc_hmlmeso:day_layer = "lower mesopelagic (3)" ;
                mnkc_hmlmeso:night_layer = "epipelagic (1)" ;
                mnkc_hmlmeso:carbon_to_wet_weight_conversion_factor = 11.9 ;
        short zeu(time, latitude, longitude) ;
                zeu:_FillValue = -32767s ;
                zeu:standard_name = "euphotic_zone_depth" ;
                zeu:units = "m" ;
                zeu:scale_factor = 0.00149805563635319 ;
                zeu:add_offset = 58.8652907365642 ;
        short npp(time, latitude, longitude) ;
                npp:_FillValue = -32767s ;
                npp:standard_name =
"net_primary_productivity_of_biomass_expressed_as_carbon_in_sea_water" ;
                npp:units = "mg m-2 day-1" ;
```

```
                npp:scale_factor = 0.230372484038343 ;
                npp:add_offset = 7548.38481551561 ;

// global attributes:
                :title = "Global ocean low and mid trophic levels biomass content hindcast" ;
                :history = "Created on 2022-06-14" ;
                :source = "SEAPODYM-LMTL 3.0.0" ;
                :references = "http://www.cls.fr; http://www.seapodym.eu" ;
                :institution = "CLS" ;
                :Conventions = "CF-1.7" ;
                :source_physical_variables = "GLOBAL_REANALYSIS_PHY_001_030 CMEMS
product" ;
                :source_biogeochemical_variables =
"OCEANCOLOUR_GLO_CHL_L4_REP_OBSERVATIONS_009_082 and
GLOBAL_REANALYSIS_BIO_001_029 CMEMS products" ;
                :net_primary_production_model = "VGPM" ;
                :date_field = "2019-12-25" ;
                :spatial_resolution = "0.083x0.083" ;
                :temporal_resolution = "1 day" ;
                :domain = "global" ;
}

For the cmems_mod_glo_bgc_my_0.083deg-lmtl-FPhys_P1D-i dataset:
netcdf cmems_mod_glo_bgc_my_0.083deg-lmtl-Fphy_P1D-i_20191224 {
dimensions:
        latitude = 2040 ;
        longitude = 4320 ;
        time = UNLIMITED ; // (1 currently)
        depth = 3 ;
variables:
        float latitude(latitude) ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        int time(time) ;
                time:standard_name = "time" ;
                time:units = "seconds since 1970-01-01 00:00:00" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
        int depth(depth) ;
                depth:standard_name = "depth" ;
                depth:units = "m" ;
                depth:axis = "Z" ;
                depth:positive = "down" ;
                depth:comment = "In fact units are pelagic layer number; 1: epipelagic layer; 2:
upper mesopelagic layer; 3: lower mesopelagic layer" ;
        short T(time, depth, latitude, longitude) ;
                T:_FillValue = -32767s ;
```

```
                T:standard_name =
"sea_water_potential_temperature_vertical_mean_over_pelagic_layer" ;
                T:units = "Celsius degree" ;
                T:scale_factor = 0.00055657182841547 ;
                T:add_offset = 15.8527576451365 ;
        short U(time, depth, latitude, longitude) ;
                U:_FillValue = -32767s ;
                U:standard_name =
"eastward_sea_water_velocity_vertical_mean_over_pelagic_layer" ;
                U:units = "m/s" ;
                U:scale_factor = 5.78025531312874e-05 ;
                U:add_offset = 0.224249370361803 ;
        short V(time, depth, latitude, longitude) ;
                V:_FillValue = -32767s ;
                V:standard_name =
"northward_sea_water_velocity_vertical_mean_over_pelagic_layer" ;
                V:units = "m/s" ;
                V:scale_factor = 5.14087418864705e-05 ;
                V:add_offset = -0.037516727518307 ;
        short pelagic_layer_depth(time, depth, latitude, longitude) ;
                pelagic_layer_depth:_FillValue = -32767s ;
                pelagic_layer_depth:standard_name =
"sea_water_pelagic_layer_bottom_depth" ;
                pelagic_layer_depth:units = "m" ;
                pelagic_layer_depth:scale_factor = 0.0150333419815309 ;
                pelagic_layer_depth:add_offset = 507.402483291177 ;

// global attributes:
                :title = "Physical forcings of global ocean low and mid trophic levels biomass
content hindcast" ;
                :source = "SEAPODYM-LMTL 3.0.0" ;
                :references = "http://www.cls.fr; http://www.seapodym.eu" ;
                :institution = "CLS" ;
                :Conventions = "CF-1.7" ;
                :source_physical_variables = "GLOBAL_REANALYSIS_PHY_001_030 CMEMS
product" ;
                :spatial_resolution = "0.083x0.083" ;
                :temporal_resolution = "1 day" ;
                :domain = "global" ;
                :date_field = "20191224" ;
                :history = "Created on 2022-06-14" ;
}
```